Anarchy of Effects? Exploring Attention to Online Advertising and Multiple Outcomes

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ABSTRACT

This research examines the relationship between attention to online advertising and brand attitude, aided recall, and purchase intention. Results indicate that attention to an ad is affected by ad type (pictorial vs. text) and the interaction between ad location (left vs. right) and page (image-oriented vs. textual), suggesting a range of factors that impact attention. Furthermore, under the online conditions of this study, attention is positively related to aided recall and to purchase intention, but negatively associated with brand attitude. Results are interpreted in the framework of dual attitude theory (Wilson, Lindsey, & Schooler, 2000) and other effects models. Although a clear "hierarchy of effects" appears to be elusive, the results suggest that marketers must fully evaluate advertising goals prior to creative development. © 2011 Wiley Periodicals, Inc.

Advertising is a huge industry, with total U.S. advertising revenue reaching $285.1 billion in 2008 (Plunkett's Almanac, 2009). Because of this large investment in advertising, marketers are continually searching for improvements in their advertising results. A growing amount of money is being spent on Internet advertising, with revenues totaling $23.4 billion in 2008 (up 10.4% over 2007; IAB, 2009). However, banner ad click-through rates are low (between 0.1 and 0.2 percent for standard ads; DoubleClick, 2007) and only 10% of business

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executives believe that banner advertising is highly effective in generating new business (Forrester Research, 2006). Despite poor click-through performance, online advertising was expected to be used by 90% of companies in 2008, up from 80% in 2005 (Richard, 2006), and banner ad revenue is expected to grow by 16.9% annually from 2006 to 2011 (Veronis Suhler Stevenson, 2007). Perhaps advertisers continue to use online ads because of "branding" benefits (Briggs & Hollis, 1997) despite lack of attention (Yoo, 2005) or recall (Fang, Singh, & Ahluwalia, 2007).

The purpose of this research is to examine the effect of online ad stimuli on attention, as well as the relationship between attention and multiple advertising outcomes such as aided recall, brand attitude, and purchase intention. This study contributes to the marketing literature by (1) examining the effect of a range of ad stimuli factors (ad type, ad location and page type) on attention; (2) evaluating the relationship between ad attention and multiple advertising outcomes [prior research has primarily focused on effects of attention on memory (Pieters, Warlop, & Wedel, 2002) or attitude (Heath, Brandt, & Nairn, 2006; Yoo, 2005]; and (3) exploring circumstances under which lower-attention processes impact advertising effects (Van Osselaer et al., 2005) by comparing dual attitude theory (Wilson, Lindsey, & Schooler, 2000) to other hierarchy of effects models.

THEORETICAL BACKGROUND

The theoretical background begins by summarizing research into what has been termed non-conscious or preattentive processing (i.e., effects upon preferences or memory from the mere exposure to stimuli). Then potential effects of attention on attitudes and other advertising outcomes are summarized in the context of dual attitude theory.

Mere Exposure Theory

Starting in the 1970s, research contradicting rational, utility-based theory began to emerge. One of the more radical non-cognitive theories was that the relationship between stimulus and affect does not necessarily depend on traditional cognitive processes (Moreland & Zajonc, 1977). Later experimental results (e.g., Kunst-Wilson & Zajonc, 1980; Moreland & Zajonc, 1979) suggested that affective judgments are independent of, and precede, cognitive operations (Zajonc, 1980). Subsequent research found similar "mere exposure" effects (e.g., Bornstein, 1989; Heath, Brandt, & Nairn, 2006; Janiszewski, 1988, 1990; Shapiro, MacInnis, & Heckler, 1997).

The evidence for low-attention effects, at least under certain conditions, now appears to be well documented. A recent review of memory, attitude, and persuasion (Johar, Maheswaran, & Peracchio, 2006) acknowledges that one of the main themes to emerge from consumer research in the past 15 years is the impact of implicit processes on consumer behavior. Furthermore, recent research has identified mere exposure effects in low-attention situations (Heath, Brandt, & Nairn, 2006; Yoo, 2005). A major remaining research issue is the extent and circumstances under which subconscious processes impact advertising effects (Van Osselaer et al., 2005).

Dual Attitude Theory

Despite substantial research into mere exposure effects on attitude, issues remain about the extent to which subconscious processes impact advertising effects (Van Osselaer et al., 2005). Research favoring strong mere exposure effects suggests that implicit attitudes are strong and enduring (Shapiro & Krishnan, 2001), that attitudes decline more in high-elaboration conditions (Chow & Luk, 2006) and that spontaneous affective reactions rather than cognitions have a greater impact on choice (when processing resources are limited; Shiv & Fedorikhin, 1999).

Pioneering research into dual attitudes (Wilson, Lindsey, & Schooler, 2000) suggests that people can have different attitudes toward the same object, both an implicit attitude (i.e., people are unaware of basis for evaluation, automatic activation) and an explicit attitude. Both the new and original attitudes remain in memory, accessible under different circumstances (Wilson, Hodges, & LaFleur, 1995). The endorsed attitude depends on whether there is cognitive capacity to retrieve the explicit attitude and "override" the implicit attitude (Wilson, Lindsey, & Schooler, 2000). Thus, proponents of dual attitude theory (Wilson, Lindsey, & Schooler, 2000) assert that automatic or implicit attitudes can guide behavior under low cognitive resources and that future research should explore how attitudes and behavior are formed under different conditions.

CONCEPTUALIZATION AND HYPOTHESES

The conceptual model addresses predicted relationships among different advertising outcomes, focusing in particular on the effect of attention.

Ad Stimuli Effects on Attention

In order to generate a range of ad attention levels, different ad stimuli (ad type, ad location, and page type) were manipulated. Eye-tracking research indicates that a pictorial stimulus is superior in capturing attention (Pieters & Wedel, 2004), suggesting the importance of ad type. Moreover, prior research indicates the importance of ad location (Dreze & Hussherr, 2003; Janiszewski, 1998) as well as the criticality of visual competition (Janiszewski, 1998) and page background (Pieters & Wedel, 2004) when measuring attention, suggesting the importance of both ad location and page type. Therefore, it is expected that the ad stimuli manipulations (pictorial vs. text ad, left vs. right location, and image-oriented page vs. textual page) will have a differentiating effect on attention.

H1: Attention will be significantly differentiated by manipulations of ad type, ad location, and page type.

Relationships Between Attention and Other Advertising Outcomes

Figure 1 illustrates a simple, graphical model based on dual attitude theory (Wilson, Lindsey, & Schooler, 2000), highlighting potential effects of attention and dual attitudes.

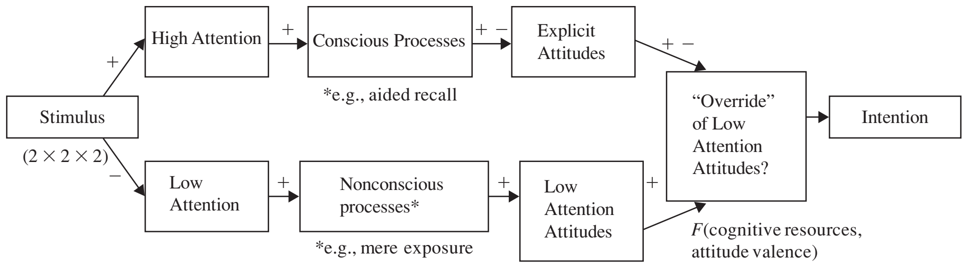

Figure 1. Simple adaptation of dual attitude model.

The hypotheses assume that, although there will be low overall attention to banner ads (given low Internet ad click-through rates), some conditions (e.g., a pictorial ad, ad location, page context, etc.) will attract greater ad attention. These differences in attention, measured by attention tracking (see Methodology section), are predicted to affect some advertising outcomes. Recent research (Yoo, 2005; Heath, Brandt, & Nairn, 2006; Goodrich, 2007), for example, suggests that higher attention appears to weaken mere exposure effects and that "brand-building" might be more effective with lower levels of ad attention.

Generally speaking, higher attention is predicted to be associated with higher aided recall and explicit effects, although lower attention is expected to produce greater positive mere exposure effects, with potential override of implicit effects prior to purchase intention. More specifically, as explained in the hypotheses which follow, it is expected that attention will be positively related to aided recall (e.g., explicit memory is limited by attention; Finlay, Marmurek, & Morton, 2005), but negatively related to attitude (e.g., disliking judgments are more cognitive, liking is more spontaneous; Herr & Page, 2004). Furthermore, attention's effect on purchase intention is expected to be mediated separately by recall (Dahlén & Lange, 2005; Gronhaug, Kvitastein, & Gronmo, 1991; Haley & Baldinger, 1991; Scheier, 2003) and by attitudes (e.g., Biehal, Stephens, & Curlo, 1992; Brown & Stayman, 1992; Fishbein & Ajzen, 1980). However, although lower attention is expected to prompt positive mere exposure effects, these effects might be overridden by study conditions (e.g., prior brand attitudes, explicit intention question, etc.). Potential direct effects of ad stimuli on outcomes have been controlled (see Appendix B).

It should be noted that the "low-attention attitudes" in this paper are more closely associated with "mere exposure" or lack of attention to the ad (see Methodology), distinct from attitude measurements from evaluative priming (Fazio et al., 1995) or implicit association tests (IAT; Greenwald, McGhee, & Schwartz, 1998). There is some interesting research regarding IAT and advertising outcomes (e.g., Friese, Wänke, & Plessner, 2006; Scarabis, Florack, & Gosejohann, 2006), but this study has a different focus. As suggested by pioneers in evaluative priming (Fazio et al., 1995), neither their priming measures nor IAT necessarily connote lack of stimulus attention or awareness (Fazio & Olson, 2003). It should also be emphasized that pictorial, or higher-attention ads, are not always associated with less favorable attitudes (e.g., Yeung & Wyer, 2004). This study, through the expected overall low attention to banner ads, explores the point of contrast between mere exposure/low attention (for which positive attitudinal effects have been shown) and slightly higher attention (e.g., with pictorial banner ads).

Attention—Aided Recall. Specifically, prior research suggests that attention and aided recall are related. For example, explicit memory is limited by attention, although implicit (non-conscious) memory is not (Finlay, Marmurek, & Morton, 2005; Shapiro & Krishnan, 2001). Furthermore, eye-tracking research suggests that attention increases information available to memory, which improves brand recall (Pieters, Warlop, & Wedel, 2002), and that attention is correlated with recall (Scheier, 2003). Therefore, one could logically expect attention to be significantly related to aided recall.

H2: Attention will have a significant positive relationship with aided recall.

Attention—Brand Attitude. Although processing of banner ads is expected to be primarily implicit, some conditions (e.g., a pictorial ad) will likely attract greater attention (Pieters & Wedel, 2004). Thus, the ad stimuli (see Methodology) are expected to yield a range of attention measures. Based on prior low-attention research, it is expected that level of attention will be negatively associated with attitude (e.g., disliking judgments are more cognitive, liking is more spontaneous; Heath, Brandt, & Nairn, 2006; Herr & Page, 2004) due to reduced mere exposure effects (Bornstein, 1989) and higher elaboration conditions (Chow & Luk, 2006). Mere exposure effects appear to be more pronounced when subjects are unaware of the exposure (Zajonc, 2001) or have low attention to the stimulus (Yoo, 2005; Heath, Brandt, & Nairn, 2006; Goodrich, 2007).Thus, higher attention is expected to be associated with lessened mere exposure effects, while low attention should activate implicit processes and more favorable attitudes.

H3: Attention will have a negative relationship with brand attitude.

Attention—Purchase Intention. Under dual attitude theory (Wilson, Lindsey, & Schooler, 2000), the attitude–behavior relationship depends on the type of attitude involved (implicit or explicit) and the type of behavior involved (implicit or explicit). If greater cognitive resources are available, people retrieve explicit attitudes from memory, which can override the implicit attitude (Wilson, Lindsey, & Schooler, 2000).

Attention has traditionally been more difficult to measure than other advertising effects such as recall, but the influence of attention on purchase intention appears to depend on factors such as brand strength, attitude valence, cognitive resources, and so on. For example, purchase intention is higher for a weak brand when participants *cannot* recall the ad (Dahlén & Lange, 2005), suggesting a mere exposure effect, but purchase intention is higher for a strong brand (e.g., top-ranked shaver, as in this study; Euromonitor, 2005) when consumers *can* recall the ad (Dahlén & Lange, 2005).

Overall, research into direct effects of attention on purchase intention appears mixed. Eye-tracking research indicates that attention is significantly related to choice and sales (Lohse, 1997; Scheier, 2003) and that reduced attention has an adverse impact on sales (Janiszewski, 1998). On the other hand, some research suggests potential dominance of implicit effects on purchase intention in "low attention" environments (e.g., Auty & Lewis, 2004; Baker, 1999; Mandel & Johnson, 2002; Shapiro, MacInnis, & Heckler, 1997). However, most research showing implicit effects on purchase intention has utilized fictitious or unfamiliar brands.

The preponderance of the evidence indicates that, for a strong brand, attention will be positively related to purchase intention.

H4: Attention will have a significant positive association with purchase intention.

Aided Recall—Purchase Intention. The effect of attention on purchase intention is potentially mediated by other advertising outcomes such as attitude and aided recall. If there are sufficient processing resources, purchase intention is expected to be positively related to explicit processes such as recall (for strong brand; Dahlén & Lange, 2005). Furthermore, strong positive correlations have been found between recognition and purchase intention (Gronhaug, Kvitastein, & Gronmo, 1991; Haley & Baldinger, 1991; Scheier, 2003), in contrast to other research that finds that recall is a poor measure of effect on purchase (e.g., Jones & Blair, 1996; Ross, 1982). Although prior research is somewhat mixed, it more strongly suggests a positive relationship between recall and purchase intention.

H5: Aided recall will have a positive relationship with purchase intention.

Aided Recall—Brand Attitude. Research generally finds low (e.g., Ross, 1982) or nonsignificant (e.g., Law & Braun, 2000; Van Reijmersdal, Neijens, & Smit, 2007) correlation between brand preference and recall. Affect can be processed at low levels of attention and does not always perform well on recall measures (Heath & Nairn, 2005; Shapiro & Spence, 2005). Traditional mere exposure research (e.g., Janiszewski, 1988, 1990) has specifically excluded participants who recognized the ad, yet still found significant attitudinal effects. Although attitude might sometimes be related to recall (e.g., Chattopadhyay & Alba, 1988), such conditions (e.g., evaluative, attribute oriented) are not expected in this lower-attention online environment.

H6: Aided recall will have no significant relationship with brand attitude.

Brand Attitude—Purchase Intention. Traditional marketing theory and prior research suggests a positive relationship between attitude and purchase intention (e.g., Brown & Stayman, 1992; Fishbein & Ajzen, 1980). Although there is disagreement about whether implicitly generated attitudes are more ephemeral (Petty, Cacioppo, & Schumann, 1983) or enduring (Chow & Luk, 2006; Naylor, Raghunathan, & Ramanathan, 2006; Winkielman, Zajonc, & Schwarz, 1997), most research shows positive direct effects of brand attitude on purchase intention (Brown & Stayman, 1992).

H7: Brand attitude will have a positive relationship with purchase intention.

Brand Attitude—Purchase Intention (under High or Low Ad Attention). There appears to be greater stability in the relationship between attitudes and behavior under high involvement than under low involvement (Petty, Cacioppo, & Schumann, 1983). Furthermore, brand memories (e.g., thoughts, feelings, experiences, images, perceptions, beliefs, and attitudes) are thought to influence consumer purchase decisions (Keller & Lehmann, 2003).

With a strong brand (Dahlén & Lange, 2005), these brand memories are expected to be positive. Thus, it is predicted that brand attitude will be positively associated with purchase intention under higher ad attention.

H8: Brand attitude will have a positive relationship with purchase intention under higher ad attention.

If banner ad attention is low, some research suggests that positive implicit effects should carry through to purchase intent (e.g., Auty & Lewis, 2004; Baker, 1999; Mandel & Johnson, 2002), with implicit effects dominating explicit effects on purchase intention. In dual attitude terms (Wilson, Lindsey, & Schooler, 2000), if attention and recall are low, explicit attitudes might not override implicit attitudes, so that positive mere exposure effects dominate.

However, other research suggests that just asking a purchase intention question activates product category memory (Fitzsimons & Morwitz, 1996) and that elaboration increases the contaminative effects of hypothetical questions on choice (Fitzsimons & Shiv, 2001). For example, people who think about reasons for their choice move away from spontaneous preferences (e.g., Wilson & Schooler, 1991). Furthermore, the presence of a specific brand (as in this study) acts as signal during a choice task, with greater activation of memory retrieval (Plassmann, Kenning, & Ahlert, 2007). For "strong" brands (Dahlén & Lange, 2005), these memories will be positive and the relationship between ad attitude and purchase intention is weaker (Brown & Stayman, 1992). Thus, the prompting of explicit effects will potentially override initial implicit (ad-generated) effects on purchase intention, although the threshold for such overrides has not yet been formally established.

H9: Brand attitude will have a negative association with purchase intention under low ad attention.

METHODOLOGY

Design

This study's methodology combined characteristics of a controlled experiment with those of a typical Internet environment. The experiment was conducted with the cooperation of a major Internet ad serving company and a company that performs attention tracking. A major Internet portal site and a large entertainment site provided home page images, and a major consumer products company (electric shaver) provided banner ads. A $2 \times 2 \times 2$ between-subjects factorial design was used, in which the type of banner (pictorial or verbal), ad location (left or right of Web page), and the type of Web page (text or image-oriented) were manipulated. Participants were randomly assigned to one of the eight ad stimuli conditions. Each stimulus was tested in a clutter of eight Web page images. Vertical (160×600 IMU "skyscraper") banners were used, since the dimensions were most applicable to right–left location on the Web page. The major portal site was targeted for the "text-oriented" page and the entertainment site for the

Figure 2. Ad stimuli examples.

"image-oriented" page (manipulation check confirmed by 21 of 23 undergraduate business students). Samples of the ad stimuli and the clutter of eight Web page images are shown in Figures 2 and 3.

Sample and Data Collection

Participants from a voluntary panel (provided by a major online survey firm) were randomly assigned to treatment groups in an online environment. The panel was recruited through an online survey firm, adhering strictly to ESOMAR codes and guidelines for access panels (www.esomar.org). No monetary reward was offered for completion, although respondents could support charities, have a chance to win prizes, and so on. The sample of eight monadic cells included at least 100 respondents per cell (total $n = 884$), with all participants at least 18 years of age. Participants were recruited over an approximately one-week period prior to the study (February 22 and February 28, 2006), and all participants resided in the United States. Potential panelists had previously been asked to join panel communities based on intrinsic interest in survey participation and sharing of opinions. All samples were randomly generated prior to survey deployment. Participants came from a wide range of backgrounds (e.g., approximately half female, with a broad range of ages, family composition, incomes, and education). A summary of participant information is shown in Table 1.

Attention was tracked online in real time using attention tracking (see overview in Measure Development), by instructing test participants to use the mouse as a pointing device. It should be noted that Web page images, not "live" Web pages, were used, so participants did not have to download graphics/pages during the survey. The complete survey, with images, was downloaded prior to implementation, so individual page/graphics download time was not expected to be a major issue. After a short training procedure, participants clicked whatever caught their eye on the series of eight Web page images, each presented for 10 seconds. Participants were instructed to click quickly (one to two times per second) and spontaneously (not to think about it first). Thus, the mouse behavior in this experimental task is somewhat different from normal mouse behavior on a Web page, and the presentation time does not necessarily mirror normal

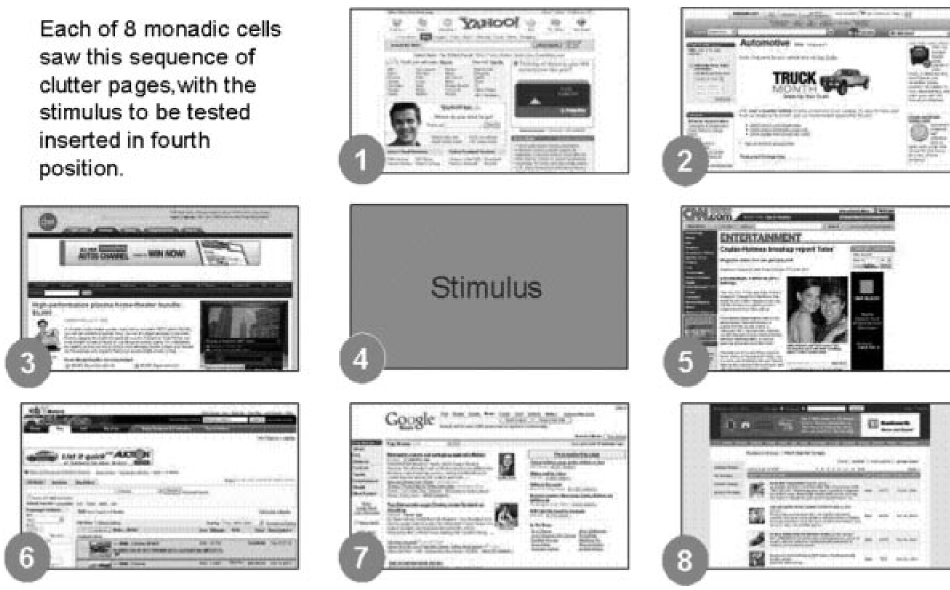

Figure 3. Clutter of eight Web pages.

Table 1. Participant Characteristics.

	Age	Gender	Education	Marital
Mean	51			
Median	53			
Percent	10% < 30, 40% 31–50, 30% 51–60, 20% > 60	55% female, 45% male	4% high school/trade, 68% college, 28% graduate/advanced	62% married, 38% unmarried

online behavior. However, the methodology allows more comparable data to be gathered from the diverse sample. Participants were then asked to complete an online questionnaire designed to measure aided recall, brand attitude, and purchase intention.

Measure Development and Validation

The attention-tracking technology used in this study exploits the fact that visuo-spatial attention generalizes across response effectors (eye, head, and hand movements; Astafiev et al., 2003) such as mouse clicks (Scheier, 2003). Empirical evidence indicates that this methodology provides a valid and reliable measure of visuo-spatial attention (Scheier, 2003). In particular, it has been shown that fixations measured with this methodology are significantly correlated ($r = 0.90$, $p < 0.01$) with fixations measured by traditional eye-tracking devices. With typical eye-tracking, fixations are inferred from continuous data streams. Attention tracking directly measures fixations and avoids reliability issues of eye-tracking measures from blinking and eye tearing (Pieters, Rosbergen, & Wedel, 1999). The internal and external validity of this attention-tracking methodology is further explained in Appendix A.

Definition of Attention and the Measurement of Other Outcomes. "Attention" is defined as the percent of clicks on a particular region (e.g., the banner ad) divided by the total number of clicks on the page, yielding a "percent attention" measure with theoretical values from 0 to 100. Brand attitude was measured using five attitude items (good, appealing, attractive, likeable, pleasant; Janiszewski, 1988, 1990) on a 7-point Likert scale (7 highest and 1 lowest). In this study brand attitude (similar to Janiszewski, 1990) was measured, which is positively related to attitude toward the ad in low-involvement situations (Droge, 1989; Vakratsas & Ambler, 1999). Aided recall was measured using a dichotomous variable (Unnava & Burnkrandt, 1991), with respondents marking recalled brands from a list of 25 companies. Purchase intention was measured with a single item (Morrison, 1979) on a 7-point Likert scale (1 highest, "very likely to purchase," and 7 lowest, "very unlikely to purchase").

RESULTS

The five items measuring attitude all loaded onto a single factor, with individual loadings above 0.80. Reliability was strong, with a Cronbach's alpha of 0.94. Attitude item scores were summed and used as an overall indicator of attitude. Table 2 displays cell means for the eight ad stimuli combinations and each advertising outcome. The cell mean measure for aided recall equals the percent of participants who correctly recalled seeing the shaver brand.

The hypothesized effects between variables were tested using the appropriate statistical methods for each predicted relationship. Depending on the relationships being considered, ANOVA adaptations (Hastak & Olson, 1989, Grewal, Monroe, & Krishnan, 1998; Campbell & Keller, 2003), logistic regression (Kumar & Krishnan, 2004), or linear regression was used. Effect sizes are reported using the following measures: Eta^2 for ANOVA/ANCOVA, b (standardized beta) for linear regression, and odds ratio [Exp(B)] for logistic regression. Note that in many cases, effect sizes are quite small, which is not uncommon in mere exposure-related research (Harrison, 1977) or research in which subtle manipulations (Prentice & Miller, 1992) are used (i.e., low-attention environments, exploratory vs. goal-oriented tasks, etc.). After hypothesis testing, additional mediation analyses were performed to shed light on potential mediating relationships (see Appendix B).

Ad Stimuli and Attention

It was predicted that the ad stimuli manipulations would significantly differentiate measured attention. An ANOVA on attention revealed a significant main effect of ad type [pictorial vs. text; $F(1,883) = 68.79$, $p < 0.001$, Eta^2 = 0.073] and a significant interaction between ad location (left–right) and page type [image or text-oriented; $F(1,883) = 13.10$, $p < 0.001$, Eta^2 = 0.015]. There were no other significant relationships. The main effect results revealed that people exposed to pictorial ads (M = 16.96) exhibited higher measured ad attention than people

Table 2. Cell Means by Page Type, Ad Placement, and Ad Type.

	Text Page				Image Page			
	Left ad		Right ad		Left ad		Right ad	
	Pictorial	Text	Pictorial	Text	Pictorial	Text	Pictorial	Text
Attention	17.2 (1.28)	11.0 (1.17)	15.2 (1.28)	7.6 (1.20)	15.8 (1.22)	8.7 (1.16)	19.8 (1.18)	11.9 (1.24)
Aided recall	0.07 (0.031)	0.18 (0.028)	0.02 (0.031)	0.12 (0.029)	0.50 (0.029)	0.14 (0.030)	0.02 (0.029)	0.02 (0.028)
Attitude	14.9 (0.70)	16.5 (0.64)	15.5 (0.70)	15.8 (0.66)	15.5 (0.67)	16.9 (0.68)	14.6 (0.65)	15.4 (0.64)
Intention (Low = Better)	4.1 (0.15)	4.6 (0.14)	4.1 (0.16)	4.4 (0.15)	3.7 (0.15)	4.8 (0.15)	4.1 (0.15)	4.2 (0.14)

Note: Standard deviations are indicated in parentheses.

exposed to text ads ($M = 9.83$), consistent with prior research (e.g., Pieters & Wedel, 2004). A Bonferroni's test to examine differences in attention resulting from the significant ad location × page type interaction revealed greater attention to rightward ads on an image-oriented page ($M = 15.8$) versus a textual page ($M = 11.2, p < 0.01$), and greater attention to rightward ($M = 15.8$) versus leftward ($M = 12.3$) ads on an image-oriented page ($p < 0.05$). All of the three advertising variables, either alone or in combination with the other variables, significantly differentiated measured attention, supporting H1.

Attention and Aided Recall

Logistic regression using attention and aided recall indicated that attention was positively related to aided recall [Wald statistic = 4.46, $df = 1, p < 0.05$, Exp(B) = 1.02], supporting H2. Thus, the higher the measured attention, the greater the aided recall, although the effect size [Exp(B)] was fairly small.

Attention and Brand Attitude

Linear regression using attention and brand attitude revealed that attention was negatively related to attitude [$F(1,882) = 7.03, p < 0.01, R^2 = 0.08, b = -0.047$], supporting H3. Higher attention to the ad, therefore, was associated with lower brand attitude, consistent with prior attention-related research (e.g., Bornstein, 1989; Heath, Brandt, & Nairn, 2006), although the effect size was quite small.

Attention and Purchase Intention

Linear regression using attention and purchase intention revealed that attention was positively related to purchase intention [$F(1,882) = 15.63, p < 0.001$, $R^2 = 0.17, b = 0.13$], supporting H4. As measured attention to the ad increased, purchase intention significantly increased.

Aided Recall and Purchase Intention

ANOVA results indicated that aided recall was not significantly associated with purchase intention [$F(1,882) = 0.77, p > 0.10$], failing to support H5.

Aided Recall and Brand Attitude

ANOVA results indicated that aided recall had no significant association with brand attitude [$F(1,882) = 0.28, p > 0.10$], supporting H6.

Brand Attitude and Purchase Intention

Linear regression of brand attitude and purchase intention revealed that attitude was *negatively* related to purchase intention [$F(1,882) = 229.35, p < 0.001$, $R^2 = 0.17$, $b = -0.45$], contrary to H7, which had predicted an overall positive relationship.

Attitude, Attention, and Purchase Intention

H8 and H9 were tested by using two groups, a low-attention group (attention < 10, which is less than chance) and a high-attention group. Linear regression revealed significantly negative relationships between attitude and intention for *both* the high-attention group [inconsistent with H8; $F(1,525) = 150.09$, $p < 0.001$, $b = -0.47$] *and* the low-attention group [consistent with H9; $F(1,353) = 75.48$, $p < 0.001$, $b = -0.42$].

DISCUSSION AND CONCLUSION

The results of this analysis suggest that, under the particular conditions of this online study, (1) variations in ad stimuli significantly differentiated measured attention, (2) there was an inverse relationship between attention and brand attitude, (3) attention was positively related to purchase intention, and (4) a clear hierarchy of effects is elusive.

The significant effect of ad type, ad location, and page type on attention supports prior research into the superiority of pictorial ads in gaining attention (Pieters & Wedel, 2004) and attention-based effects of ad location and page type. However, the significantly negative effect of attention on attitude is inconsistent with a conclusion that advertisements failing to gain consumers' attention cannot be effective (Wedel & Pieters, 2000) and supports prior research into the positive nature of low-attention attitudes (e.g., Heath, Brandt, & Nairn, 2006).

Under the conditions of this study, attention had direct associations with aided recall and purchase intention, supporting more "cognitive" effects, but was negatively related to brand attitude, consistent with mere exposure theory. There was little support for mediation by aided recall or attitude of the effects of attention on purchase intention, with an insignificant effect of aided recall on intention and a negative effect of attitude on intention. The strong positive relationship of attention with purchase intention appeared to dominate other effects.

Although a clear "hierarchy of effects" appears to be elusive in this online study, the results suggest that marketers must fully evaluate advertising goals prior to creative development. For example, higher attention (e.g., with pictorial ads) and immediate purchase might be preferable for strong brands (Dahlén & Lange, 2005). However, if the goal is to slowly build long-term attitudes (perhaps for a newer brand, to avoid counterarguing by loyal users of a strong brand), then ads generating lower attention but potentially higher attitudes might be favored. Prior research has recommended low-attention brand building (e.g., Heath, Brandt, & Nairn, 2006) and multiple advertising exposures (Dahlén, 2001) for unfamiliar brands, despite lack of immediate tangible effects.

Interpretation of Effects Models

If put in dual attitudes (Wilson, Lindsey, & Schooler, 2000) terms, explicit factors in this study (e.g., strong brand, positive valence, purchase intention "cognitive trigger") appeared to override low-attention, ad-related effects on purchase intention. However, contrary to expectations, effects of both lower-attended and more highly attended online ads appeared to be overridden. Perhaps most online ads, as suggested by low click-through rates, are low-attention or "implicit" enough to be overridden by other situational effects (e.g., strong brand, purchase intention question, etc.). However, are there any better competing explanations to dual attitude theory for these results?

A general cognitive hierarchy of effects model, such as *attention–recognition–attitude–intention*, with "awareness" affecting attitudes and intentions (e.g., ELM—Petty, Cacioppo, & Schumann, 1983; anchoring and adjustment—Lynch, Marmorstein, & Weigold, 1988; MPAA—Cohen & Reed, 2006; and less-supported single path theories), was not strongly supported. Not only was there no relationship between aided recall and either attitude or intention, but most cognitive theories do not incorporate a negative relationship between attention and attitude (as does mere exposure theory). Similarly, a pure cognitive approach (e.g., *attention–recognition–intention*, that consumer preferences are rational and based on brand attributes; Stigler, 1961; Telser, 1964) was not supported, since there was not a significant recall-to-intention relationship. Another pure cognitive model (e.g., *attention-to-intention*) was not supported because attitude partially mediated the effect of attention [a chi-square difference test indicated a better model fit *with* attitude as a partial mediator ($\Delta\chi^2(2) = 204.4$, $p < 0.001$), using a structural model without the dichotomous aided recall variable].

Furthermore, a pure affective model, such as *low attention–implicit attitudes–intention*, under which consumer liking preferences (e.g., from "mere exposure") carry over to purchase intention, was not supported because there was a *negative* relationship between implicit attitudes (from mere exposure or low ad attention) and purchase intention. Thus, none of these prior models are fully supported by the results. A model with both attention and attitude included as independent predictors of purchase intention is more consistent with the data.

One conclusion from the data is that a clear hierarchy of advertising effects is elusive. Some researchers argue that there is little support for any advertising effects hierarchy, and that advertising effects should be studied in a space, with affect, cognition, and experience as three dimensions (Vakratsas & Ambler, 1999). It is possible, for example, that purchase intention (e.g., goals, experience) had a reverse-causal effect on attention in this study. A recent meta-analysis on the attitude–behavior relationship finds an inverse relation between direct experience and the attitude–behavior correlation (unless behavioral relevance and one-sidedness of the attitudinal bases are controlled; Glasman & Albarracín, 2006). Future research should address such issues to better understand the interrelationship of advertising effects.

Limitations and Future Research

It should be noted that the results of this study were performed in a specific type of online environment, and that the results may not necessarily be replicable to other environments. For example, Web page images were used rather than "live" Web pages, allowing greater control of page exposure time (10 seconds per page) and page sequence. Future research might use live Web pages to better understand visitor navigation and clickstreams. Furthermore, the choice of ad (i.e., electric shaver) may have had some effect on respondent motivations, since product relevance (e.g., Darley & Lim, 1992) and an advertisement's fit with a page (e.g., Karson & Fisher, 2005) can be an issue. Although the "exploratory" orientation of the study (with participants provided no motivation to search for specific items) likely reduced the importance of product relevance (Janiszewski, 1998), future research should utilize different advertised products and page contexts to further understand these effects in an online environment.

It should also be noted that effect sizes of significant relationships were often quite low. However, large effects from experiments generally might not be expected from experimental manipulations (Fern & Monroe, 1996) that are subtle (Prentice & Miller, 1992), such as experiments regarding mere exposure (Harrison, 1977). Furthermore, small effects might be considered more important when results can generalize to large numbers of people (LaTour, 1981) and when the effort to change stimuli is minimal, as can be the case in online advertising.

Although this study focused primarily on attention and other advertising effects, some ad stimuli had direct effects on outcomes such as aided recall and purchase intention (see Appendix B). Future research could focus more specifically on such direct effects of ad stimuli.

Conclusion

With online ad spending in the billions of dollars, the Internet is one of the largest advertising communication media. This study has attempted to provide further insight into the relationship between attention and other advertising outcomes, so that the results can potentially be used to better match online advertising design/placement with desired advertising goals.

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APPENDIX A

Internal and External Validity of Attention-Tracking Methodology

Internal Validity. The correspondence of eye tracking and clicking data was found by accident during an eye-movement study (Egner, Itti, & Scheier, 2000), in which pointing, eye movement, and clicking data using multiple image types were compared to a neural network model of visual attention (e.g., Itti & Koch, 2000; Egner, Itti, & Scheier, 2000; Scheier et al., 1999). Although the model predictions correlated only moderately with behavior, there was a strong and significant overlap between the eye tracking and the click data, implying that the mouse can be used as a means to measure attention. A summary of different aspects of the attention-tracking method's validity is provided below, but additional information is available from the author on request.

Neurological research generally supports the validity of this study's attention-tracking methodology. For example, brain activities of human participants and monkeys were mapped in three conditions: (1) attending, (2) looking, and (3) pointing to a peripheral visual location (Astafiev et al., 2003). Regions relevant for attentional processing, such as the human frontal eye field (FEF) and a region in parietal cortex called LIP, were similarly recruited in all conditions, "suggesting an attentional role that generalizes across response effectors" (Astafiev et al., 2003). Thus, whether we attend, look, or point at a location in space, the same brain areas are involved. Similarly, psychophysical research indicates that attention for various sensor and motor modalities is commonly controlled (in the brain's superior colliculus; Krauzlis, Liston, & Carello, 2004; Kustov & Robinson, 1996; Lane et al., 1973; Sparks, 1999), consistent with a premotor theory of attention (e.g., Clark, 1999; Rizzolatti et al., 1987). More specifically, a gaze-anchoring process locks onto the pointing target until the pointing movement is completed (Neggers & Bekkering, 2001). The brain enforces ocular fixation of a pointing target, or inhibits secondary saccades away from a target, until pointing is completed ("gaze anchoring"). Thus, goals for eye movements as well as mouse clicks are selected by the same mechanism, consistent with cognitive psychology research into selection-for-action (Allport, 1987; Neumann, 1987), in which attention is an auxiliary function of action, not vision.

A typical eye-tracking device records eye movements at a high temporal and spatial resolution. The goal is to find out which areas people focus on most strongly and which areas are ignored. The human brain only consciously processes information when the eyes rest for more than a few hundred milliseconds, with these resting periods called "fixations." The human eye re-fixates about two to four times a second. While the eye is in motion ("saccades"), there is very little information processing. Therefore, the focus of analysis in eye-tracking experiments is on the fixations, which reflect fixations of attention.

One key issue in typical eye tracking is the inference of fixations from the continuous data stream delivered by the eye-tracking device. The attention-tracking technology used in this study directly measures fixations by using a pointing device (the computer mouse) to infer eye fixations and attention, not by tracking actual eye movements. With this study's attention-tracking methodology, attention focuses the gaze and the hand on the same location, so that attention location can be measured by mouse click position. The resulting fixation data are equivalent to attention data gathered by means of eye-tracking fixations. In particular, attention measures from Web site stimuli, using traditional eye-tracking and this study's attention-tracking methodology, are significantly correlated (Scheier, 2003). In a double-blind study (Scheier, 2003), a set of 10 Web sites was tested, with each home page shown for 10 seconds and fixations defined to be stationary phases of the eye of 200 milliseconds and longer. The average correlation between clicks and eye fixations across all Web sites and regions tested was 0.90 ($p < 0.01$). Furthermore, reliability might be improved by the direct measurement of fixations and because eye tracking measures can be influenced by subjects' excessive blinking or tear fluid (Pieters, Rosbergen, & Wedel, 1999).

External Validity. Often, visual field effects are studied using a procedure that (1) forces a person to fixate on a point, (2) presents the target information outside the foveal viewing area (i.e., the area associated with looking at this point), and (3) removes the target information before attention can shift to the information. This procedure ensures that any effect on a critical dependent measure is a consequence of parafoveal viewing (a visual field effect) instead of foveal viewing (an attention effect). Similarly, some research on visual field asymmetries (e.g., Hagenbeek & Van Strien, 2002) presents stimuli tachistoscopically for brief moments in the various visual fields, with exposures too brief to allow eye fixations and a fixation-cross ensuring fixation in the middle of the screen.

This study's procedure does not control the initial point of fixation, but allows attention to fixate on Web page stimuli, including the ads under investigation. Thus, this research is focused more on the way ad attention is influenced by ad characteristics and page placement than on a methodologically exacting examination of visual field effects. This study tests the advertising effects of the horizontal positioning of a stimulus relative to its context (i.e., on the left or right part of a page), though not strictly controlling retina sensitivity and the visual brain for information on the left and right of a fixation point. As mentioned, standard Web images, ads, and settings were used, enhancing external validity. Page layout and participant goal effects were controlled by using different page types and by randomly sampling participants who were provided with identical instructions.

APPENDIX B

Tests for Mediating Relationships among Advertising Outcomes

Tests for mediating relationships primarily utilized mediation analysis (Baron & Kenny, 1986). Ad stimuli main effects and interactions were also evaluated and are noted if significant.

Attention–Aided Recall. First, an ANOVA revealed a significant main effect of ad type [$F(1{,}883) = 68.79, p < 0.001$, Eta$^2 = 0.073$] and an ad location $\times$ page type interaction [$F(1{,}883) = 13.10, p < 0.001$, Eta$^2 = 0.015$] on attention. Second, logistic regression (Kumar & Krishnan, 2004) of ad stimuli on aided recall indicated a main effect of ad type [Wald statistic = 5.81, $df = 1$, $p < 0.05$, Exp(B) = 3.01], an ad location $\times$ page type interaction [Wald statistic = 4.53, $df = 1, p < 0.05$, Exp(B) = 6.11], and an ad type $\times$ page type interaction [Wald statistic = 26.41, $df = 1, p < 0.001$, Exp(B) = 0.054]. Third, logistic regression with attention included as a covariate yielded a significant effect of ad type [Wald statistic = 7.38, $df = 1, p < 0.01$, Exp(B) = 3.52], an ad location $\times$ page type interaction [Wald statistic = 0.60, $df = 1, p < 0.05$, Exp(B) = 7.69], and an ad type $\times$ page type interaction [Wald statistic = 26.32, $df = 1$, $p < 0.001$, Exp(B) = 0.053], as well as a significant negative effect of attention [Wald statistic = 8.23, $df = 1, p < 0.01$, Exp(B) = 0.976]. Thus, when the ad stimuli and attention are both included in the regression equation, attention had a negative relationship with aided recall [due to Exp(B) < 1]. Interestingly, when only attention is regressed on aided recall without the ad stimuli included, the expected positive relationship between attention and aided recall occurs [Wald statistic = 4.46, $df = 1, p < 0.05$, Exp(B) = 1.015], consistent with H2.

Attention–Brand Attitude. First, ANOVA results of ad stimuli on attention revealed a significant main effect of ad type [$F(1{,}883) = 68.79, p < 0.001$, Eta$^2 = 0.073$] and a significant ad location $\times$ page type interaction [$F(1{,}883) = 13.10, p < 0.001$, Eta$^2 = 0.015$]. Second, only ad type significantly predicted attitude, with pictorial ads ($M = 15.3$) associated with less favorable attitudes than text ads ($M = 16.1$) [$F(1{,}882) = 5.05, p < 0.05$, Eta$^2 = 0.006$]. Third, ANCOVA revealed a significant *negative* relationship between attention [$F(1{,}882) = 4.53$, $p < 0.05$, Eta$^2 = 0.005$] and attitude, and the effect of ad type was not significant [$F(1{,}882) = 1.38, p > 0.10$], indicating that attention completely mediated the influence of ad type on attitude.

Attention–Purchase Intention. First, as noted, there were significant main effects of ad type and the ad location $\times$ page type interaction on attention. Second, ANOVA revealed a main effect of ad type [$F(1{,}883) = 19.35, p < 0.001$, Eta$^2 = 0.022$] as well as the ad location $\times$ ad type interaction [$F(1{,}883) = 6.01$, $p < 0.05$, Eta$^2 = 0.007$] on purchase intention. Third, ANCOVA revealed a significant positive relationship between attention [$F(1{,}882) = 4.53, p < 0.01$, Eta$^2 = 0.010$] and purchase intention, as well as significant effects of ad type [$F(1{,}882) = 11.96, p < 0.01$, Eta$^2 = 0.013$] and the ad location $\times$ ad type interaction [$F(1,882) = 6.37, p < 0.05, \text{Eta}^2 = 0.007$]. Thus, attention had a significantly positive direct effect on purchase intention, partially mediating the effect of ad type (but not the ad location × ad type interaction) on purchase intention.

Aided Recall–Purchase Intention. First, logistic regression indicated that attention positively affected aided recall [Wald statistic = 4.46, $df = 1$, $p < 0.05$, $\text{Exp}(B) = 1.02$]. Second, attention positively predicted purchase intention [$F(1,882) = 15.62, p < 0.001, b = 0.13$]. Third, however, ANCOVA shows that aided recall did *not* predict purchase intention with attention as a covariate [$F(1,882) = 0.365, p > 0.10$], nor was aided recall related to purchase intention without attention included. Attention appeared to have a direct relationship with purchase intention independent of aided recall.

Brand Attitude–Purchase Intention. Linear regression results indicated that (1) attention (*negatively*) predicted attitude [$F(1,882) = 7.03$, $p < 0.01$, $b = -0.047$], (2) attention (*positively*) predicted purchase intention [$F(1,882) = 15.62, p < 0.001, b = 0.13$], and (3) attitude (*negatively*) predicted purchase intention [$t(1,881) = 14.88, p < 0.001, b = -0.45$] when attention is also a predictor, technically satisfying requirements for partial mediation; the effect of attention on purchase intention remained significant [$t(1,881) = 3.07$, $p < 0.01$, $b = 0.092$] but decreased when attitude was included. However, the effect of attitude on purchase intention was significantly *negative*, opposite from expectations.